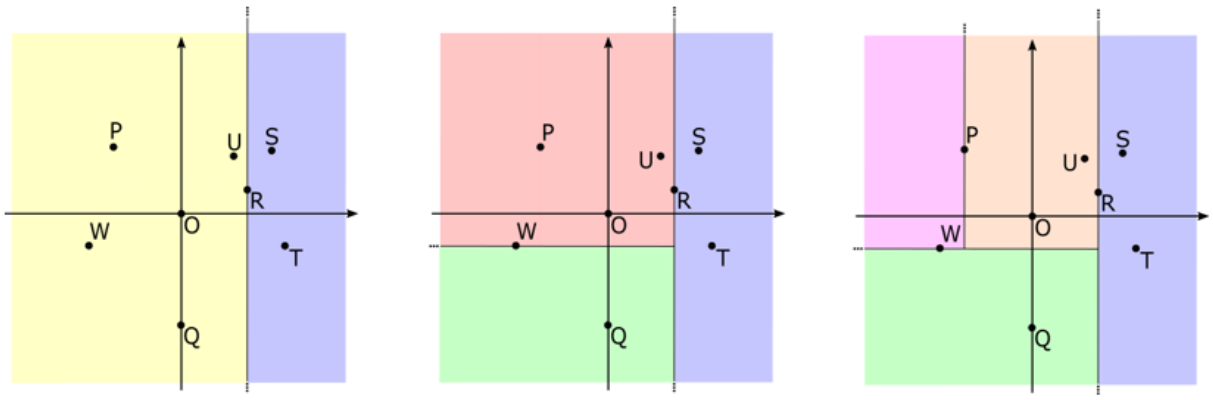


Figure 9.2 Partitioning points in the 2-D Cartesian space by cycling through the directions along which we split. For the first split (left) we choose point `R` and draw a vertical line (parallel to `y` axis, `x` coordinate is constant) passing through it. We have such created 2 half-spaces, on the left (yellow) and right (blue) of this line, grouping points `W`, `P`, `O`, `Q`, `U` on one side, and `S`, `T` on the other. Point `R` is the pivot of this partitioning.
Next, we choose point `W` in the yellow partition: this time, we draw a horizontal line (parallel to `x` axis, `y` coordinate is constant): it splits the yellow partition into 2 new partitions, one in the top-left area of the plane (red), containing `P`, `O` and `U`, and one in the bottom-left area (green), with just `Q`.
If we further split the red area at point `P`, we need to use again a vertical line, as shown in the right-most part of the figure.

Notice that in the Cartesian plane a vertical line through a point `P=(Px,Py)` has a peculiar characteristic: it is parallel to the y axis, and all points on the line have the same value for their x coordinate, `Px`. Likewise, a horizontal line passing